

Xinyu Jiang

Robotics & Embodied AI

Hangzhou Dianzi University · Intelligent Science and Technology · Hangzhou, China

23061740@hdu.edu.cn · jiaoxiangyue3@gmail.com · github.com/Jxy-yxJ · linkedin.com/in/xinyu-jiang-a00b17402 · Jxy-yxJ.github.io/en/

EDUCATION

B.S. in Intelligent Science and Technology — Hangzhou Dianzi University

2023 – 2027

Focus: robotics, computer vision, machine learning, optimization.

RESEARCH PROJECTS

RoboECG — autonomous robotic ECG electrode placement

2026

- A UR3 perceives a supine patient's chest from an overhead depth camera, locates the six precordial electrodes (V1–V6) from clinical rules, presses them onto the skin, and self-checks — in Isaac Sim.
- 6/6 first-attempt placements with 0.14–0.36 mm contact error; depth-based localization 6.08 mm mean over 360 held-out placements (95% CI 5.53–6.64).
- Validated the localization rules against 25 statistical-shape torsos and 120 measured electrodes (PhysioNet/CinC 2007).

Robotic Thyroid Scanning in Isaac Sim

2026

- Reproduced the coarse-localization stage of an autonomous thyroid-ultrasound system: RGB-D perception (MediaPipe), depth-fused target localization, and collision-aware UR3 planning.
- 0.70 mm reach error with ≥ 2.26 cm guaranteed body clearance; depth fusion improves localization 25% (6.23 $\rightarrow$ 4.67 cm) across 7 perturbations; 40 simulator-free unit tests + CI.

MemoryGuard — bounded active memory maintenance

2026

- Long-horizon embodied agents act on stale remembered locations; MemoryGuard decides when to verify, verifies with a grounded signal, refreshes on staleness, then acts, under a bounded revisit budget.
- Built on AI2-THOR with a Grounded-SAM2 detector and a vision-language model; found that a strong VLM detects staleness 6/6 but updates correctly only 3/6.

SKILLS

Programming: Python, Git / GitHub Actions, Linux

Robotics simulation: NVIDIA Isaac Sim, AI2-THOR, UR3 + Lula IK

Perception & ML: RGB-D geometry, MediaPipe, OpenCV, PyTorch, Transformers, NumPy

Web: Astro, TypeScript

ACTIVITIES

Photographer; drummer in the indie band 巧克力文件岛.